Course Policy and Norms

Course: AELL101 – Introduction to Electrical Engineering

Instructor: Dr. Ankit Singhal, sankit@iitdabudhabi.ac.ae
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Office Hours: Thursday 10 AM – 11 AM, and Tutorial sessions by appointment

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Course Objectives: To enable students

- to understand broader principles of electrical engineering
- to identify and analyze electrical elements and create circuits

Text Book:

1. "Circuits, Devices and Systems – A first course in Electrical Engineering," by Ralph J Smith and Richard C Dorf, Wiley, 5th edition.

Topics:

1. Network Theorems (KCL, KVL, Thevenin)
2. RLC circuits
3. Phasor analysis of AC circuits. Single phase and 3-phase circuits
4. Operational amplifiers
5. Introduction to Digital circuits
6. Magnetic circuits, Transformers: Modeling and analysis
7. Energy in magnetic field, Electromechanical energy conversion principle

Assessment:

Mid-term	25
Major	30
Quizzes (best 2 out of 3)	2 x 10 = 20
Tutorial+assignments	12
Group Activities	13
Total	**100**

1. There will be 4 quizzes: The timings and venues will be declared before these quizzes. The best 2 out of 4 will be considered. NO further make-up quiz will be conducted in any circumstance.
2. Please use tutorial sessions for clearing any conceptual as well as numerical doubts from the tutorial instructors.
3. Grading will be relative.
4. Group activities (total 13 marks) may include both in-class and out-of-class tasks as follows:
    - Lecture scribe: 5 marks: round 1 (3) + round 2 (2)
    - Whole group presence in Exams: 2 marks: MidSem (1) + EndSem (1)
    - Whole group above 30% marks in: MidSem (1 mark) + EndSem (1 mark)
    - Group project: 4 marks

Lectures Plan

S.N.	Topic	Hours	Resources from text book	Tutorial
1	Basic electrical elements and quantities	3	Chapter 1	Tut 1
2	Network theorems (KCL, KVL, Thevening, Superposition, Maximum Power)	3	Chapter 2	Tut 2
Quiz 1 (1 hour)				
3	Natural response of first order circuits (RC, RL): source free circuits	3	Chapter 4.1, 4.2	Tut 3
4	Natural response of second order circuits (RLC): source free circuits	3	Chapter 4.3, https://simufisica.com/en/rlc-circuit/	Tut 4
5	Complete step response of first and second order circuits (RL, RC, RLC)	2	Book not recommended, use class notes and solved example document	Tut 4
Mid Semester Exam				
6	AC waveforms, phasor analysis of AC circuits, AC impedance	2	Chapter 5.3	Tut 5
7	Steady-state AC power calculation	2	Chapter 7	Tut 5
8	Resonance in AC circuits	2	Chapter 7	Tut 6
9	3-phase AC circuits and power	3	Chapter 7	Tut 7
Quiz 2 (1 hour)				
10	Diode circuits and rectifiers	3	Chapter 3.7, 3.8, 11	Tut 8
11	Operational Amplifiers (OpAmps)	3	Chapter 3.6, 16	Tut 9
12	Introduction to digital circuits	6	Chapter 13	Tut 10, 11
Quiz 3 (1 hour)				
13	Magnetic circuits and transfornmers	3	Chapter 20,21	Tut 12
14	Electromechanic energy conversion: Machines	3	Chapter 22	Tut 12
15	Introduction to systems	3	Chapter 10	
Quiz 4 (1 hour) + Final Exam				